SMU SINGAPORE MANAGEMENT UNIVERSITY | School of Computing and Information Systems

Academic Year 2026-27 Term 1 Project
AI-Driven Software Engineering (SE201)
Version 1.00

Synopsis

Generative AI is changing how software is conceived, built, and maintained. This project lets students collaborate on a focused project centred on large language models and practical software engineering problems. By identifying where LLMs can meaningfully support the software development lifecycle and applying techniques such as retrieval-augmented generation, agentic workflows, and tool integration, students will design, build, and evaluate a working AI-enabled solution. The solution may improve an existing product or practice, or prototype a new use case made possible by LLMs. Throughout the project, students will disclose how they use AI and critically assess their solution's limitations, risks, and ethical implications.
An LLM will be the centrepiece of your project, whether you revise an existing product or process, or create something novel. While GUIs and cloud-hosted models make the project more polished, a command-line interface and a locally run model are perfectly acceptable and will not count against your grade. Architecture, data flow, a rational use of AI to solve a problem, and clear metrics for measuring success are the key features of a successful project.

Overall Project Assessment

Assessment Component	Due Date	Total	Team	Individual
Proposal	2 October (Friday) at 23:59	**5%**	5%	
Checkpoint	18 October (Sunday) at 23:59	**5%**	5%	
Presentation / Deliverables	10 November (Tuesday) at **12:00**	**20%**	15%	5%
Peer Evaluation	17 November (Tuesday) at 23:59	**5%**		5%
Total		**35%**	**25%**	**10%**

The presentation and deliverables are assessed by the instructor using Appendix A. The individual presentation is assessed by clarity of speech and ability to answer queries and convey key messages. The peer evaluation is assessed by your team members.

You are allowed to use AI tools as part of your development process. Your team must track use of these tools as part of its AI use disclosure, which will be described in a separate document.

Project Proposal (5%)

Each team is required to submit a **two-page** proposal document by **2 October (Friday) at 23:59**. Submission will be via eLearn / Assignments / SE201 Project Proposal.

I will grade your submission on how it addresses the following points. The proposal shall include:

1. Team and project information
   - Give the project title, team name, and each member's name and area of responsibility.
2. Problem statement
   - Describe the software / software engineering problem or new use case your project addresses.
   - Identify the intended users or stakeholders and explain why the problem matters to them.
   - State whether your project improves an existing product or practice, or prototypes a new use case made possible by AI.
3. Justification for using AI
   - Explain why AI is a suitable approach to this problem.
   - Briefly describe a conventional, non-AI alternative, and explain what AI offers that it does not.
4. Proposed solution and technical approach
   - Describe the main features of your solution. Include a high-level architecture diagram showing the main components and how data flows between them.
   - State which techniques you will apply (for example retrieval-augmented generation, agentic workflows, or tool integration), and justify each choice.
   - List the models, APIs, frameworks, and data sources you plan to use, and confirm that you can access them.
5. Evaluation plan
   - How will you measure whether your solution works, using metrics, test cases, or a baseline for comparison?
   - Describe the test data or scenarios you will use.
   - State what result would count as success.
6. Checkpoint targets
   - State the specific deliverables you commit to showing at the prototype checkpoint. At a minimum, include:
     - One working core feature
     - Implemented architecture: the main components in place and connected, even if they are simplified or stubbed.

Project Checkpoint (5%)

Each team is required to submit a checkpoint document and supporting materials by **18 October (Sunday) at 23:59**. Submission will be via eLearn / Assignments / SE201 Project Checkpoint.

Your submission will be assessed on how it addresses the following points, but I am mostly checking that your team is making progress on your project. I do not expect you to achieve all your goals, and you will possibly discover unforeseen problems. This is valuable information. Document this as part of your checkpoint and your project.

The checkpoint shall include:

1. Read access to a shared code repository that includes:
   a. A README with setup instructions
   b. A commit history showing contributions from every team member
2. Show your main architecture components in place and connected. You may simplify or stub a component, but you must document which ones.
3. Demonstrate at least one core feature as described in your proposal.

4. Show that your chosen AI technique (for example retrieval-augmented generation, a workflow, or tool integration) is implemented and working in the prototype in some limited fashion.
5. A document
    a. Results from at least one round of testing against the metrics or test cases in your proposal and describe any failures you observed.
    b. A progress report against the checkpoint targets you committed to in your proposal and explain any targets you did not meet.
    c. Describe any changes to scope, design, or technology since the proposal, and give your reasons.
    d. Your AI use disclosure log to date.

Project Presentation and Deliverables (20%)

Submit all deliverables in a zipped file to the eLearn / Assignments / Project Submission before the start of class on **10 November (Tuesday) at 12:00**.

There is no formal report document. The presentation will be the primary way to describe your project. Make sure the slides are detailed enough to be read without supporting speech or use speaker notes to give supplementary information.

Use the supporting documents to submit additional test results or any other relevant information.

Deliverables include:

- Presentation slides
- Code repository links or a repository snapshot
    - README with setup and runnable examples
- Supporting documents
    - Test results or sample run logs
    - Information cut from your presentation due to time constraints
    - *Optional* videos demonstrating the project's features
- AI use disclosure log

Each team will be given 20 minutes (15 minutes for presentation and 5 minutes for Q&A) to present in class during week 13.

All members must attend and present without electronic aids, such as phones, presenter notes on the laptop, etc. You may read from written or printed notes.

All members should be prepared to answer questions.

The presentation should include the following (there can be some variation based on the scope of your project):

1. **Project title and team details**
2. **Problem identification and justification for using AI**
   Define a clear software engineering problem or new use case. Explain why AI is a suitable approach, and compare it with a conventional, non-AI solution.
3. **Technical design and implementation**
   Show the high-level view of how your design works and how the components of your architecture work together to make a useful product. Do not show code unless it is essential. What model did you select and why?

4. **Evaluation and evidence of effectiveness**
   What datasets did you use and what tests did you devise to demonstrate that the design works? What did you learn from your successes and failures?
5. **Responsible AI use and disclosure**
   Assess your solution's risks, including hallucination, security, privacy, bias, and intellectual property, and explain how you address them. What did you discover about the AI tools used to develop your project? Any lessons learned or insights that can be applied to future work?
6. **Conclusion**
   What did you achieve? Are there parts of your proposal that were unfinished due to time or technical limitations?

Within-Team Peer Evaluation (5%)

*Please raise **major** team issues to your instructor's attention early and do not wait until the peer evaluation to express your concerns.*

After project submission, each team member must assess their other team members using the peer evaluation tool on eLearn, by **17 November 2026 (Tuesday) 23:59.** A penalty will be given if you fail to complete the peer evaluation.

The evaluations are on a scale of **1 to 10** and include the following categories.

- Team player
- Communication skill
- Quality of work
- Commitment
- Contribution

Guidelines

- Please be frank when evaluating your peers.
- Take the evaluation seriously. You should not give everyone in the team the exact same score.

APPENDIX A Assessment Rubrics

Table A1: Assessment criteria for the team component

Assessment Criteria	Emerging	Progressing	Competent	Accomplished
Problem Identification and Rational use of AI	Minimal analysis of the need or benefits of a specific AI model over prior designs for the selected use case.	Provides a basic analysis of the need or benefits for a specific model, but presented evidence not directly linked to selected use case.	Provides a good analysis of the need or benefits for a specific model with relevant evidence but may lack strong connection to project goals.	Defines a clear software / software engineering problem or new use case. Explains why a specific AI model is a suitable approach and compares it with a conventional solution.
Technical Design and Implementation	Prototype is missing or non-functional, or very poorly developed, showing little-to-no relevance to stated problem.	Prototype is basic and functional but lacks major features. Relevance to stated problem is superficial.	Functional prototype with most features in basic form, demonstrating relevance to stated problem.	Functional, well-featured prototype demonstrating strong relevance to stated problem.
Evaluation and Evidence of Effectiveness	Missing or poorly selected evaluation criteria with little to no metrics.	Metrics / dataset selected with limited justification or connection with selected task. Limited results and analysis.	Metrics / dataset carefully selected with demonstrated connection to selected task. Some metrics demonstrating effectiveness and limited analysis on how the software works and where it fails.	Metrics / datasets carefully selected and relevant to selected task. Detailed results demonstrating effectiveness. Well-developed analysis on how the software performed and where / how it goes wrong.
Responsible AI use and Disclosure	Missing or incomplete disclosure. Missing or incomplete AI risk assessment.	Partial AI use disclosure and basic risk assessment with some mitigations.	Complete AI use disclosure and risk assessment. Effective mitigations.	Comprehensive AI use disclosure and risk assessment, with extensive mitigations.
Team Collaboration	Poor coordination within the team; tasks are not effectively completed.	Team collaborates but struggles with coordination; some tasks are incomplete or late.	Team collaborates well; all tasks are completed on time.	Team demonstrates excellent collaboration; all deliverables are of high quality and submitted on time.

Table A2: Assessment criteria for the individual component

Assessment Criteria	Emerging	Progressing	Competent	Accomplished
Individual Presentation Skills	Presentation is unclear, disorganised, and difficult to follow; speaker shows little confidence or engagement.	Presentation is somewhat clear but lacks organisation or coherence; speaker shows some confidence but may struggle with delivery.	Presentation is clear, well-organised, and delivered with confidence; speaker engages the audience effectively.	Presentation is clear, engaging, well-organised, and delivered with strong confidence; speaker captivates the audience and communicates ideas persuasively.
Individual Ability to Answer Questions	Unable to answer questions or provides unclear, incomplete answers; lacks understanding of the topic.	Struggles with answering questions; provides basic but incomplete answers, showing some understanding.	Answers most questions effectively with clear and accurate responses; demonstrates good understanding of the topic.	Answers all questions confidently, thoroughly, and accurately; demonstrates deep understanding and rationale on the solution proposed.